

AMMAR

Machine Learning / AI

Decision Tree — Heart Failure Death Event Prediction

Objective

To build a Decision Tree classifier to predict the Death Event (mortality) of heart failure patients using clinical features. The model was trained in Python and evaluated using accuracy, ROC-AUC, and feature importance.

Dataset Overview

- Dataset: Heart Failure Clinical Records (UCI / Kaggle)
- Total Samples: 299 patients | Test Set: ~60 samples
- Target Variable: DEATH_EVENT (0 = Survived, 1 = Died)
- Features: 12 clinical variables including age, serum creatinine, ejection fraction, serum sodium, platelets, etc.

Code

```
# =====
# 3. Handle Missing values
# =====
imputer = SimpleImputer(strategy='mean')
X = mydata.drop('DEATH_EVENT', axis=1)
y = mydata['DEATH_EVENT']

X = pd.DataFrame(imputer.fit_transform(X), columns=X.columns)

# =====
# 4. Train-Test Split
# =====
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=123, stratify=y
)

# =====
# 5. Train Decision Tree Model
# =====
tree_model = DecisionTreeClassifier(
    criterion='gini',
    max_depth=None,
    ccp_alpha=0.01, # pruning parameter (like cp in R)
    random_state=123
)

tree_model.fit(X_train, y_train)

# =====
```

Model Performance

Metric	Value	Metric	Value
Accuracy	83.3%	AUC Score	0.843
Precision	~81%	Recall	~79%

Decision Tree Visualization

The tree splits first on time ≤ 73.5 (follow-up period), then on serum_creatinine and ejection_fraction. Blue nodes represent survived (class 0) and orange nodes represent death (class 1).

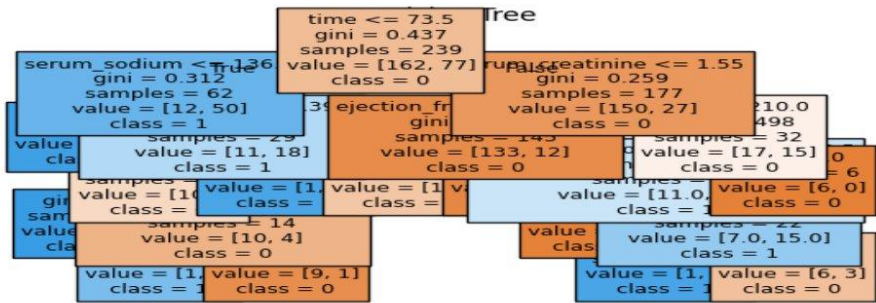
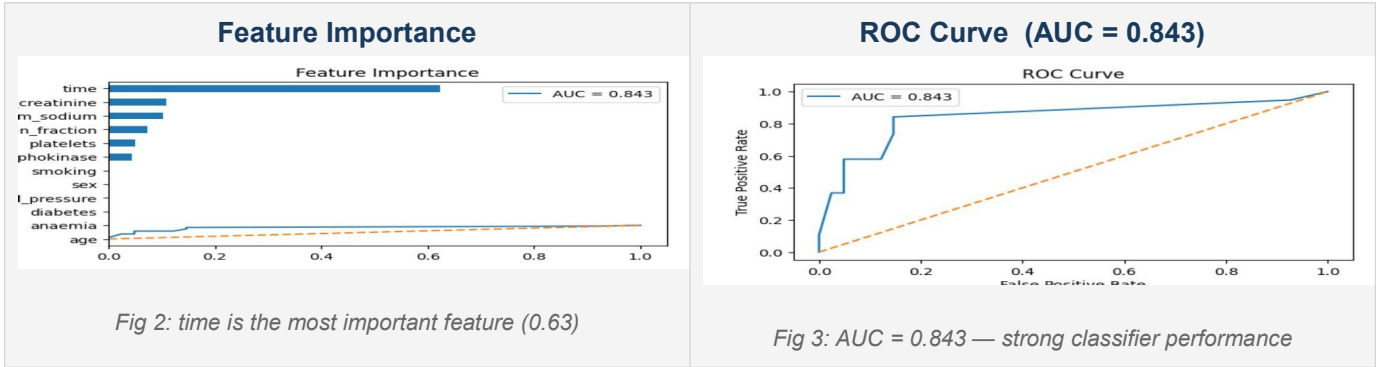


Figure 1: Decision Tree — Heart Failure Death Event (Root: time ≤ 73.5)



Key Findings

- time (follow-up period) is by far the most important predictor of death — importance score 0.63.
- serum_creatinine and serum_sodium are the next most influential clinical features.
- AUC = 0.843 indicates the model has strong discriminating ability between survivors and non-survivors.
- The Decision Tree is interpretable — clinicians can follow the branches to understand predictions.

Conclusion: The Decision Tree model achieved 83.3% accuracy and AUC of 0.843 in predicting heart failure death events. The follow-up time and serum creatinine level are the strongest clinical predictors of mortality.

— End of Assignment | Submitted by: Ammar —